BJT Current Source Experiment 1

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Pre-LAB Work

1. For the circuit given in Fig. 1, find and expression for the current flowing through the R7 resistor and for the output resistance Ro of the circuit. Ignore the Early Effect and assume that all the transistors are operating in active mode. (Hint: To calculate the current you do not need to know the value of the R7 resistor. To calculate Ro, you better review the common base configuration.)

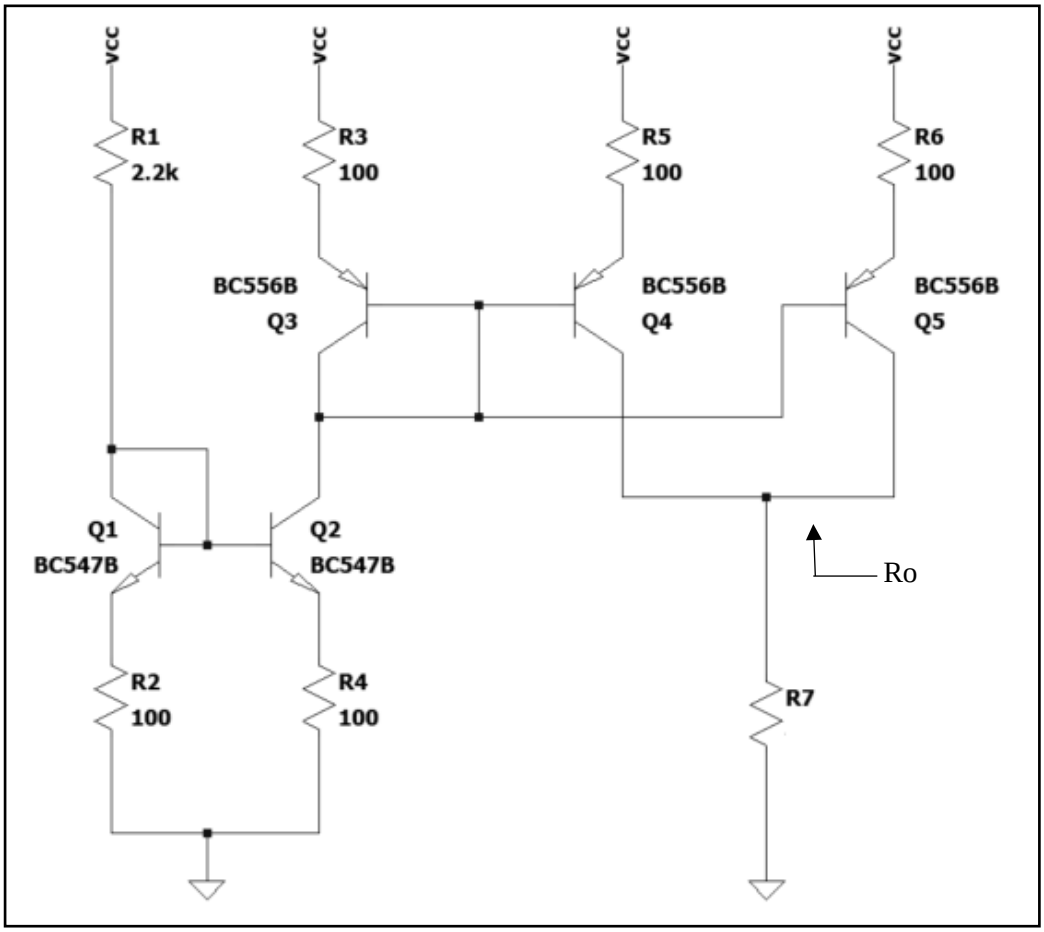
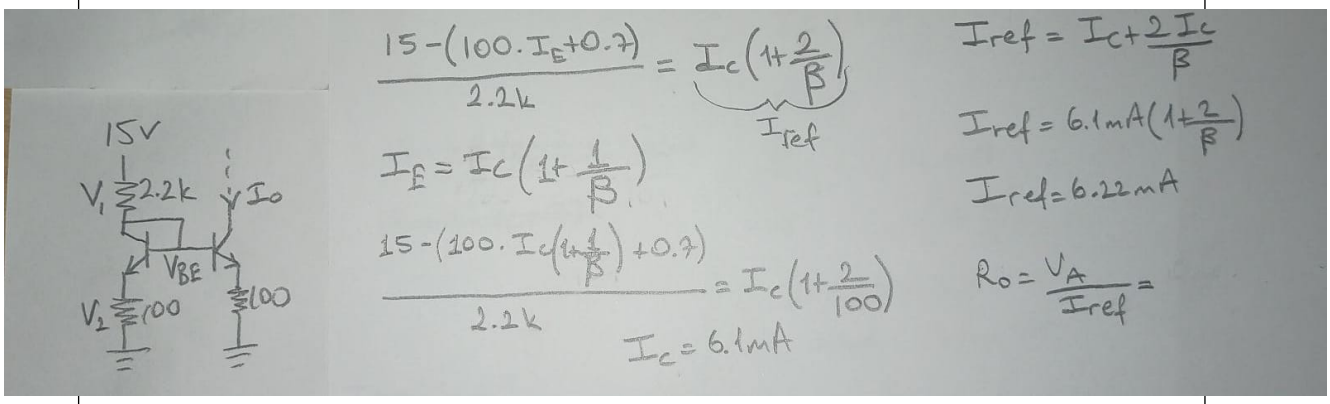


Figure 1

Answer:



2. Let Vcc be 15V and simulate the circuit in Fig. 1 with SPICE for the R7 values listed in Table 1. Measure the voltage across the R7 resistor and the current through it and fill the table.

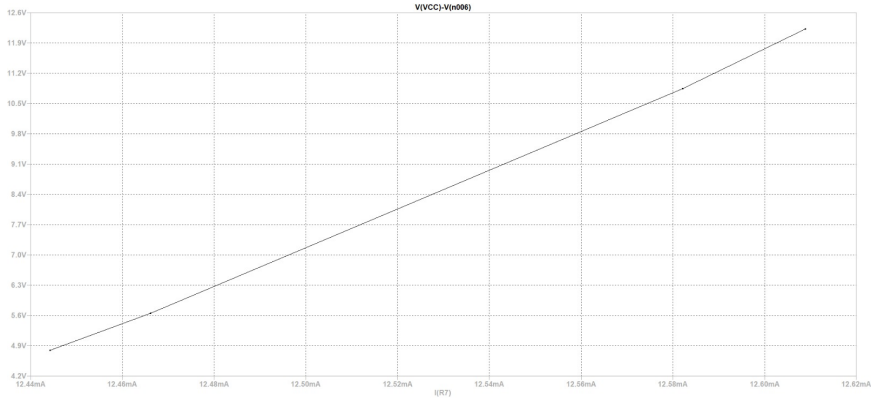
Table 1: Simulation Results

R7	V (R7)	Ix = I (R7)	Vy = Vcc – |V (R7)|
220Ω	2.773V	12.608mA	12.226V
330Ω	4.158V	12.582mA	10.841V
750Ω	9.358V	12.466mA	5.641V
820Ω	10.204V	12.443mA	4.795V
2.2kΩ	13.994V	7.378mA	1.005V

3. Based on the simulation results in Table 1 (ignore the last row) plot Ix (x-axis) vs Vy (y-axis) and fit a line. Calculate the slope of the line which will be the output resistance Ro.

Plot:

$$R_o = \frac{V_y}{I_x} = \frac{(12.226 - 4.795)V}{(12.608 - 12.443)mA} = \frac{7.431V}{0.165mA} \cong 45k\Omega$$



Equipment/Component List

Table 3: Equipment/Component List

Equipment/Component	Model/Value	Count
Breadboard	-	1
Resistor	2.2kΩ	**2**
Resistor	100Ω	**5**
Resistor	220Ω	1
Resistor	330Ω	1
Resistor	750Ω	1
Resistor	820Ω	1
BJT (npn)	BC547	**2**
BJT (pnp)	BC556	**3**
Multi-meter	-	1
Voltage Source	15V	1
Probes & Cables	-	-

Experiment Procedure

4. Set-up the circuit in Fig. 1 with the resistor values on the Fig. 1. Adjust Vcc to 15V. Values for the R7 resistor are given in Table 2. Measure the exact values of R7 resistors before connecting them to the circuit and fill the second column of the table. Connect the R7 resistor to the circuit, measure the voltage across it and fill the table.

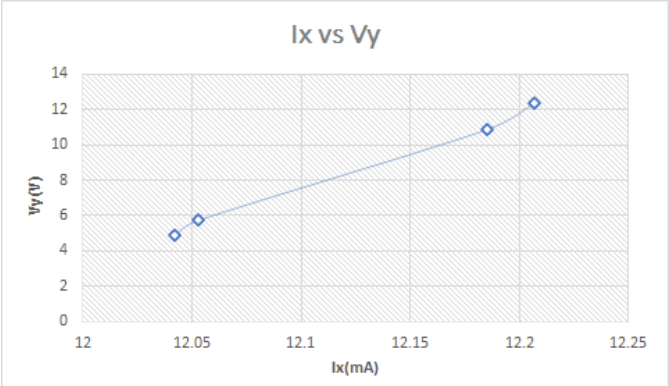
Table 2

R7	Measured (R7)	V (R7)	Ix = |V (R7)| / Meas. (R7)	Vy = Vcc – |V (R7)|
220Ω	217Ω	2.671V	12.207mA	12.379V
330Ω	329Ω	4.161V	12.185mA	10.839V
750Ω	748Ω	9.261V	12.053mA	5.739V
820Ω	817Ω	10.133V	12.042mA	4.867V
2.2kΩ	2.19Ω	13.888V	7.106mA	1.120V

5. Based on the data in Table 2 (ignore the last row), plot Ix (x-axis) vs Vy (y-axis) and fit a line. Calculate the slope of the line which will be the output resistance Ro.

Plot:

$$R_o = \frac{V_y}{I_x} = \frac{(12.379 - 4.867)V}{(12.207 \quad - \quad 12.042)mA} = \frac{7.512V}{0.165mA} \cong 45.5k\Omega$$



6. Compare the measured quantities in Table 2 with your simulation results in Table 1 and calculations in part 1. When R7 resistor is set to 2.2k the current significantly changes. Comment on the reason of this change.

Answer:

Our experiment data and calculations data results are close. The Iref value we found in the prelab is 6mA. The current on R7 is equal to 2Iref. This is about 12mA. But this condition is valid when the transistor is in the active region. When R7 resistor is 2.2K ohm, the current significantly changes because the transistor voltage difference became less than 0.7V and gets saturation.

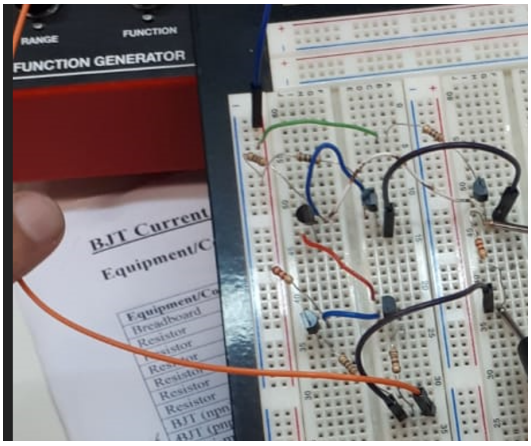


Figure 1: Constructed circuit in the lab.

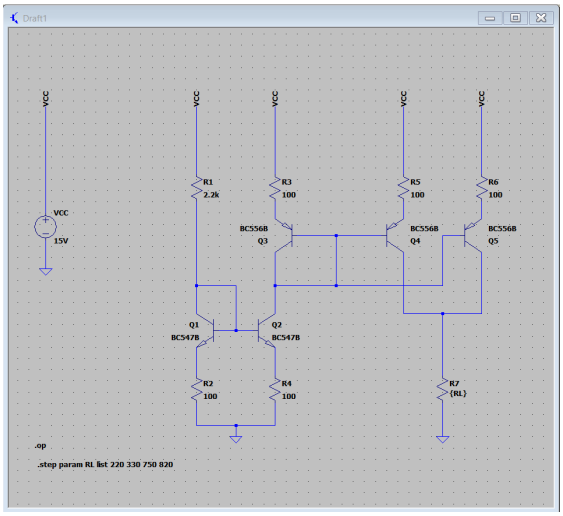


Figure 2: Constructed circuit in Ltspice.